Solutions

1.

(a)(Application of Theory. Computed example)

ø1= 45 ø2=90 as per the block diagram, and:

Local oscillator :

$LOA = 2\cos(\omega_L t - ø1)$ $LOB = 2\cos\omega_L t$

RF signal :

$A\cos(\omega_A t + ø_A) + B\cos(\omega_B t + ø_B)$

where $\omega_A = (\omega_L - \omega_{IF})$ is the wanted signal, and $\omega_B = (\omega_L + \omega_{IF})$ is the image.

<u>After mixing</u> { Recall $2\cos X\cos Y = \cos(X-Y) + \cos(X+Y)$ and $\cos(-X)=\cos(X)$ }

• $IF1 = 2A\cos(\omega_A t + ø_A)\cos(\omega_L t - ø1) + 2B\cos(\omega_B t + ø_B)\cos(\omega_L t - ø1)$

$= A\cos((\omega_L - \omega_A)t - ø1 - ø_A) + A\cos((\omega_L + \omega_A)t - ø1 + ø_A)$

$+ B\cos((\omega_B - \omega_L)t + ø_B + ø1) + B\cos((\omega_B + \omega_L)t + ø_B - ø1)$

• $IF2 = 2A\cos(\omega_A t + ø_A)\cos\omega_L t + 2B\cos(\omega_B t + ø_B)\cos\omega_L t$

$= A\cos((\omega_L - \omega_A)t - ø_A) + A\cos((\omega_L + \omega_A)t + ø_A)$

$+ B\cos((\omega_B - \omega_L)t + ø_B) + B\cos((\omega_B + \omega_L)t + ø_B)$

<u>Lowpass filter removes sum components:</u>

• $IF1 = A\cos((\omega_L - \omega_A)t - ø_A - ø1) + B\cos((\omega_B - \omega_L)t + ø_B + ø1)$

$= A\cos(\omega_{IF}t - ø_A - ø1) + B\cos(\omega_{IF}t + ø_B + ø1)$

• IF2 = A cos ($(\omega_L - \omega_A)t - \phi_A$) + B cos ($(\omega_B - \omega_L)t + \phi_B$)

= A cos ($\omega_{IF}t - \phi_A$) + B cos ($\omega_{IF}t + \phi_B$)

<u>After phase shift – ø2:</u>

• IF1 = A cos ($\omega_{IF}t - \phi_A - \phi1 - \phi2$) + B cos ($\omega_{IF}t + \phi_B + \phi1 - \phi2$)

• IF2 = A cos ($\omega_{IF}t - \phi_A$) + B cos ($\omega_{IF}t + \phi_B$)

<u>Adding signals IF1 and IF2 :</u>

IFOut = A cos ($\omega_{IF}t - \phi_A - \phi1 - \phi2$) + A cos ($\omega_{IF}t - \phi_A$)

+ B cos ($\omega_{IF}t + \phi_B + \phi1 - \phi2$) + B cos ($\omega_{IF}t + \phi_B$)

$$= 2A \cos\left(\frac{\phi1+\phi2}{2}\right) \cos\left(\omega_{IF} - \phi_A - \frac{\phi1+\phi2}{2}\right) + 2B \cos\left(\frac{\phi1-\phi2}{2}\right) \cos\left(\omega_{IF} - \phi_B + \frac{\phi1-\phi2}{2}\right)$$

(b) (Application of theory)

This should have been an image reject receiver, but it has not been properly designed because the phase shift blocks should have chosen so that the image is rejected, which is not the case.

(c) (Application of theory)

Change the values of the phase shifts so that the second term in the general equation in (a) (prior to value substitution) is eliminated. This can be achieved by making $\cos\frac{\phi1-\phi2}{2} = 0$ i.e. $\frac{\phi1-\phi2}{2} = \frac{(2n+1)\pi}{2}$ e.g. ø1 = 90° ø2 = -90°

(d) (Theory)

Oscillators harmonics can lead to spurious responses.

2.

(a) (Application of theory)

Reduction in effective bandwidth, potential instability, increase in area.

(b) (Application of theory)

Total harmonic distortion.

(c) (Application of theory)

No, it doesn't because it is a cross-coupled configuration that eliminates common mode dependencies, provided that the devices are fully matched.

(d) (Application of theory)

Two symmetrical transistors in the design that should have been identical are not.

(e) (Application of theory)

No, it is not. This is a transconductor for which the resistor helps to linearize the transconductance provided that it is much higher than the inverse of the transconductance of the transistors. But it is still a linear approximation, not large signal.

(f) (Theory)

$$Vin = Vbe1 + ReIc1 - Vbe2 - ReIc2$$

$$= Ic1(re1 + Re) - Ic2(re2 + Re)$$

where the values of re1 and re2 depend on the instantaneous value of Vin. If Re >> re1, re2 then the non-linear variation of gm with Vin is swamped:

$$Vin = Ic1Re - Ic2Re = (Ic1 - Ic2)\,Re$$

if $Iout = (Ic1 - Ic2)$; $Iout/Vin = 1/Re$

3.

(a) (New theory)

It is not possible to say without knowing the sizes of the transistors since in principle all of them are important contributors to the noise current (i.e the overall noise current is the sum of the noise currents of all of them)

(b) (Application of theory)

Since this is very low frequency the Flicker noise will dominate and since the top transistors have a very large area, the input transistor will be dominant. Hence the students should answer this question giving the expression for the Flicker noise of the input transistor as in the lecture notes.

(c) (Application of theory)

Since it is high frequency the Flicker noise will be negligible. And since the output is a current the dominant current thermal noise will correspond to the input transistor considering the large (W/L)

(d) (New theory)

Any transconductor from the lecture notes that has got a transconductance linearly dependent on a voltage or a current that could be used as a second input would lead to a valid answer (or any other transconductor the students propose)

(e) (New theory)

The third order harmonic, the reason being that in a fully differential configuration even order harmonics mathematically disappear, and odd order ones will scale down with the factorial of its order. Hence the dominant one will be the third.

4.

(a) (New theory)

Students are supposed to connect a capacitor joining the two current branches (or any alternative configuration, such as capacitors to ground) and bias the circuit, with for example two PMOS transistors with their gates connected to a constant voltage or a current mirror).

(b) (New theory)

Speed. Amplifiers have feedback and they need to have an open loop bandwidth significantly higher than the bandwidth of the input signal in order to prevent instability.

(c) (New theory)

Use PMOS devices to prevent the body effect which could cause distortion.

(d) (Theory)

Assuming all devices have equal ß:

$$Id1 - Id2 = \beta\, Vd(Vc1 - Vth) \qquad Id3 - Id4 = \beta Vd\,(Vc2 - Vth)$$

where $Vc1 = \frac{Vgs1 + Vgs2}{2}$ and $Vc2 = \frac{Vgs3 + Vgs4}{2}$

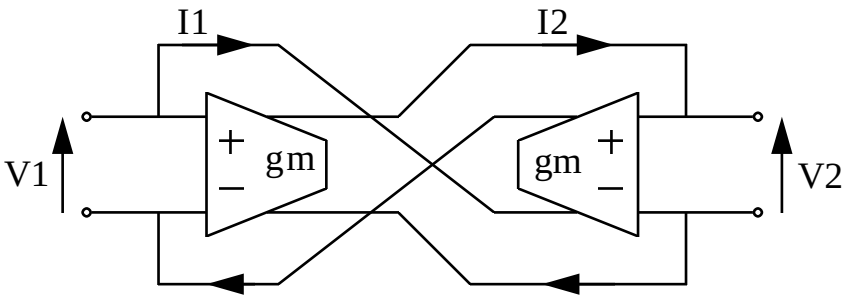
Thus $Iout = (Id1 - Id2) - (Id3 - Id4)$

$$= \beta\, Vd\,(Vc1 - Vc2) = \beta\, Vd\,(Vb - Vss)$$

(e) (New theory)

Using split capacitors (i.e. the capacitor is split in two of half of its value one is connected with its top plate to the positive branch and the bottom plate to the negative and the other capacitor the other way around).

(f) (Theory)



[3]